NinjaRS in MadLoop

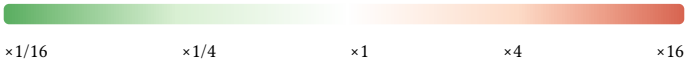
Fortran integration · scalar masters · measured benchmarks

LAYOUT FIXTURE · NOT FINAL MEASUREMENTS · AMD EPYC 9754 128-Core Processor

Four reduction libraries, identical inputs. Each coloured card gives the median complete evaluation time in **µs per phase-space point**, with its multiplier against **original MadLoop + Ninja**. The new NinjaRS interface uses oneloopmaster scalar integrals. All candidate cards use the same MadLoop source and input stream.

QCD CORRECTIONS · LOOP_SM

Process	CutTools	Ninja	Collier	NinjaRS
$u\bar{u} \to Z$ 2 ⟶ 1 · 100 points qcd-1	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point
$gg \to t\bar{t}$ 2 ⟶ 2 · 100 points qcd-2	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point
$u\bar{u} \to Zgg$ 2 ⟶ 3 · 100 points qcd-3	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point
$u\bar{u} \to ZZgg$ 2 ⟶ 4 · 100 points qcd-4	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point



Logarithmic scale · lower is faster · white is parity · red is slower

Snapshot: 0 / 32 timing cards complete; 0 / 8 correctness cases passed; 0 / 8 performance gates passed. Missing or incomplete observations remain pending. Measurements and validation are reported separately; an empty card is never treated as a speedup. Policy v2 validates the corrected model arithmetic. All 21 pairings among the seven corrected targets are required. Original-source discrepancies remain visible and the runtime denominator remains untouched original MadLoop with Ninja.

ELECTROWEAK CORRECTIONS · LOOP_QCD_QED_SM_GMU

Complete evaluation · electroweak controls

Real masses and zero widths; all generated virtual correction orders are retained. Full model names and MG5aMC process commands are listed with the reproducibility record.

Process	CutTools	Ninja	Collier	NinjaRS
$u\bar{d} \to W^+$ 2 ⟶ 1 · 100 points ew-1	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point
$e^+e^- \to \mu^+\mu^-$ 2 ⟶ 2 · 100 points ew-2	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point
$e^+e^- \to t\bar{t}\gamma$ 2 ⟶ 3 · 100 points ew-3	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ 2 ⟶ 4 · 100 points ew-4	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point	µs / point **900 000** **×0.90** vs original Ninja reference 1 000 000 µs **ML init + warm 0.500 s** 7 retained repetitions MAD 0.00 µs / point



Logarithmic scale · lower is faster · white is parity · red is slower

Timing boundary. Timed calls include the MadLoop numerator, loop reduction, scalar integrals, assembly and normal stability checks or rescues, together with per-point scale/coupling updates and return-code counting. Output is written after the timed region. “ML init + warm” begins at the first MadLoop evaluation and includes helicity initialization and one complete additional point-stream pass. It excludes parameter-card/model initialization, input reading and allocation.

At least 7 complete repetitions are required before a card receives a timing. Calibration samples are excluded. Each repetition traverses the same 100 distinct input points enough times to target a one-second timed interval. Native backend caches remain part of normal execution.

VALIDATION · SAME INPUTS, EVERY COEFFICIENT

Numerical agreement

Process	Points	Orders	Checks	Max. error / bound	Status
$u\bar{u} \to Z$	2	1	168.0	0.0000	passed
$u\bar{d} \to W^+$	2	1	168.0	0.0000	passed
$gg \to t\bar{t}$	2	1	168.0	0.0000	passed
$e^+e^- \to \mu^+\mu^-$	2	1	168.0	0.0000	passed
$u\bar{u} \to Zgg$	2	1	168.0	0.0000	passed
$e^+e^- \to t\bar{t}\gamma$	2	1	168.0	0.0000	passed
$u\bar{u} \to ZZgg$	2	1	168.0	0.0000	passed
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	2	1	168.0	0.0000	passed

Every point and squared coupling order contributes Born, finite, single-pole and double-pole values. All 21 pairings among four corrected candidate backends and three original-code backends with the verified model correction are tested. "Orders" includes the total order slot. The reported maximum is the ratio below, where S is the largest absolute coefficient at that point across all orders in the corrected cohort. Agreement requires this ratio ≤ 1. Exactly zero pairs contribute zero.

$$\frac{|a-b|}{10^{-7}\max(|a|,|b|)+10^{-10}S}$$

Performance acceptance · NinjaRS / original Ninja

Process	Repetitions	Median ratio	95% interval	Decision
$u\bar{u} \to Z$	7	0.900	0.900 – 0.900	passed
$u\bar{d} \to W^+$	7	0.900	0.900 – 0.900	passed
$gg \to t\bar{t}$	7	0.900	0.900 – 0.900	passed
$e^+e^- \to \mu^+\mu^-$	7	0.900	0.900 – 0.900	passed
$u\bar{u} \to Zgg$	7	0.900	0.900 – 0.900	passed
$e^+e^- \to t\bar{t}\gamma$	7	0.900	0.900 – 0.900	passed
$u\bar{u} \to ZZgg$	7	0.900	0.900 – 0.900	passed
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	7	0.900	0.900 – 0.900	passed

Acceptance threshold: ×1.05. The harness resamples paired run indices 10,000 times and forms a percentile 95% interval for the ratio of medians. A case passes when the upper bound is ≤ 1.05 it fails when the lower bound is > 1.05. An interval crossing the threshold is **unresolved**, not a pass. Each pair uses the same CPU, points and parameter card; measurement order alternates between repetitions.

Original-source numerical discrepancies are retained. Corrected-original references change only the verified model helper and internal-parameter postimages, rebuild their model archive, and retain original MadLoop reduction code and libraries. Their timings never replace original Ninja in the performance denominator.

MODEL PRECISION · ORIGINAL DISCREPANCIES RETAINED

Original-source numerical comparisons

Process	Pairings	Mismatches	Original comparisons
$u\bar{u} \to Z$	7	14.00	failed
$u\bar{d} \to W^+$	7	14.00	failed
$gg \to t\bar{t}$	7	14.00	failed
$e^+e^- \to \mu^+\mu^-$	7	14.00	failed
$u\bar{u} \to Zgg$	7	14.00	failed
$e^+e^- \to t\bar{t}\gamma$	7	14.00	failed
$u\bar{u} \to ZZgg$	7	14.00	failed
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	7	14.00	failed

Seven pairings compare original CutTools, Ninja and Collier to their corrected references and candidate counterparts, plus original Ninja to candidate NinjaRS. Mismatches count point/order/coefficient records for each pairing; all records remain in the dataset and snapshot. Every original numerical discrepancy remains recorded; runtime and status failures remain disqualifying. These diagnostics are distinct from the mandatory 21 corrected-cohort pairings.

Why separate source roles are necessary. The original MP model generator contains default-kind casts and a mistyped continuation constant. Compiled controls and independent higher-precision checks establish the exact corrections. A corrected result is assessed against independent corrected-original reduction paths; matching an erroneous original value is not an acceptance requirement.

Verified correction and source separation

Fresh CLI generation proves each complete model-source delta. Corrected-original references are cloned from archived original exports and receive only the exact verified model postimages. All other generated source, reduction dependencies and DHELAS libraries remain original. Their model objects, archive and subprocess are rebuilt explicitly. Candidate exports additionally contain the NinjaRS interface and validated Fortran optimization, with separate source identities.

A refreshed representative proof covers all three roles, fresh and reused workers, all four DP backends, genuine QP rescue, loop-induced amplitudes and mixed coupling orders. Missing references, invalid statuses, changed cards, missing points or stale executables prevent publication. Runtime acceptance still requires the original Ninja denominator, alternating paired order, at least seven repetitions and a resolved 95% interval within the same 1.05 threshold.

PREPARATION · SEPARATE WALL CLOCKS

Generation, compilation and initialization

Process	Original gen.	Original build	Candidate gen.	Candidate build	NinjaRS ML init + warm
$u\bar{u} \to Z$	2.00	1.00 C	3.00	1.00 C	0.500
$u\bar{d} \to W^+$	2.00	1.00 C	3.00	1.00 C	0.500
$gg \to t\bar{t}$	2.00	1.00 C	3.00	1.00 C	0.500
$e^+e^- \to \mu^+\mu^-$	2.00	1.00 C	3.00	1.00 C	0.500
$u\bar{u} \to Zgg$	2.00	1.00 C	3.00	1.00 C	0.500
$e^+e^- \to t\bar{t}\gamma$	2.00	1.00 C	3.00	1.00 C	0.500
$u\bar{u} \to ZZgg$	2.00	1.00 C	3.00	1.00 C	0.500
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	2.00	1.00 C	3.00	1.00 C	0.500

All preparation values are seconds. Generation is the recorded CLI model import, process generation and standalone export invocation; it can include library work performed by export. Build is the retained `make -j4 check` invocation: **C** begins with no subprocess object files; **I** is incremental; **?** was not classified. A clean subprocess build does not include rebuilding external libraries. ML initialization plus warming is the median of the completed run preparations beginning at their first MadLoop call; parameter-card/model setup is excluded. Generation is pending when its retained identity does not match the measured build. Generation and compilation have no performance acceptance gate.

Measurement method

The process list is fixed in `cases.json`. For two-to-many scattering, the harness constructs deterministic RAMBO-derived on-shell momenta at 1000 GeV, including the generated external masses. For $2 \longrightarrow 1$, a physical on-shell resonance is produced with distinct three-dimensional boosts. Momentum conservation and masses are checked before measurements; each case uses a different deterministic seed.

The same retained text point stream and parameter-card SHA-256 hashes are required across reducers. All decay widths are set to zero. Each point sets the renormalization scale to its partonic centre-of-mass energy. Model parameters and couplings otherwise come from the generated card.

Each executable run selects one reduction library. The driver forces stability checks, allows normal rescue within that backend, and records every timed return code. The harness rejects initialization codes, unchecked estimates, failures and unexpected backend identities. It rebuilds helicity filters before each invocation. The timing baseline is the untouched original source revision recorded on the final page. Each corrected-original reference is separately identified and compared against all four candidate backends. The common Collier scalar accuracy is 10^{-12} and the MadLoop stability threshold remains 10^{-9}.

The process is pinned to one CPU with `taskset`; OpenMP and OpenBLAS thread counts are one. Generation may use four build jobs. Timing uses Fortran `SYSTEM_CLOCK`, and includes the driver's scale updates and histogram increment. Calibration, text input, numerical output and initialization are outside the warm interval. No separate scalar-master or numerator-stage timing is inferred from total runtime.

Limits of interpretation. Confidence intervals quantify the retained run-to-run samples on this host; they do not establish performance on other CPUs or workloads. A shared host can introduce noise. Finite random streams are a regression test, not a proof over all phase space. Pending correctness, rescue or timing results remain outstanding requirements.

STABILITY · ACTUAL BACKEND USE

Timed double- and higher-precision evaluations

Process	CutTools	Ninja	Collier	NinjaRS
$u\bar{u} \to Z$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$u\bar{d} \to W^+$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$gg \to t\bar{t}$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$e^+e^- \to \mu^+\mu^-$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$u\bar{u} \to Zgg$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$e^+e^- \to t\bar{t}\gamma$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$u\bar{u} \to ZZgg$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP	28.00 DP 0.00 QP

Counts sum all retained complete timed repetitions and include repeated traversals of the point stream. They classify final successful evaluation status, rather than counting individual internal loop-reduction attempts. Full histograms are retained in `report-data.json` and the original measurement JSON. A pending entry means no accepted run is present.

Backward-compatible status interpretation

Backend	Precision	Hundreds / units digits
CutTools	DP / QP	$H = 2, U = 1$ / $H = 3, U = 9$
Ninja	DP / QP	$H = 2, U = 6$ / $H = 3, U = 8$
Collier	DP	$H = 2, U = 7$
NinjaRS	DP / QP	$H = 2, U = 8$ / $H = 3, U = 2$

The tens digit must identify a completed stability check (1 or 2). Old return integers are unchanged: the hundreds digit distinguishes Ninja QP from NinjaRS DP and PJFry++ DP from NinjaRS QP. Dedicated NinjaRS counters preserve legacy statistics interpretation.

Integration audit

The Fortran wrapper preserves MadLoop's momentum ordering, odd-rank sign reversal, thresholded invariants, Laurent ordering and normalization. Compiled recording-backend tests exercise both amplitude and interference paths, REAL16 input beyond binary64 precision, failures and nonfinite output, card parsing, old return codes, and cleanup with or without the installed module.

The bridge audit corrected the generic higher-precision auxiliary sampling scale to match Ninja's invariant/mass scale. Scalar tests cover spacelike massless and massive bubble orderings and reuse of coefficients, masses and scale. Higher-precision transport carries actual binary128 input into 256-bit arithmetic. Context cleanup joins the retained worker. See `audit.md` for scope, evidence and remaining limits.

PROFILING · SAMPLED CPU WORK

Where evaluation time is spent

Diagnostic samples, separate from timing acceptance. Profiles run a copy of an exact measured worker after initialization and warming. Sampling includes profiler overhead and output serialization between complete stream traversals. Percentages are CPU-clock sample shares, not exclusive stage timers. They are never used to decide the five-percent performance gate.

$u\bar{u} \to Z$ · original Ninja

100 points · 90 000 evaluations · 3 blocks · 405.0 samples · CPU 4

Observed work	Samples	Share
Fortran numerator · DP	194.0	47.9%
Other / unclassified	211.0	52.1%

$u\bar{u} \to Zgg$ · candidate NinjaRS

100 points · 90 000 evaluations · 3 blocks · 1000 samples · CPU 4

Observed work	Samples	Share
Fortran numerator · DP	100.0	10.0%
Fortran numerator · QP	100.0	10.0%
Native DP tensor expansion	100.0	10.0%
Compensated DP tensor expansion	100.0	10.0%
Higher-precision tensor expansion	100.0	10.0%
Higher-precision scalar masters	100.0	10.0%
Higher-precision cut geometry	100.0	10.0%
Higher-precision reduction	100.0	10.0%
Numeric transport	100.0	10.0%
Other / unclassified	100.0	10.0%

Each sample uses the first recognized stack owner or ancestor; unmatched work stays “other”. In particular, the scalar-master category covers higher-precision NinjaRS masters only. Every recorded block must reproduce all point/order coefficients, accuracy estimates and statuses of its bound measurement exactly. The report reader rechecks source/runtime identities, cards, retained raw outputs, raw recording hashes and stack-category counts. Full paths, hashes and usage counts are in `report-data.json`.

REPRODUCIBILITY · PINNED SOURCES AND RETAINED INPUTS

Source and toolchain record

Component	Recorded revision / version
Original MadLoop	27147b3300f556a75661a886e92bdbe00558ee25
NinjaRS	0d0d7e7a8033431b60a8580731084742e0655cda
oneloopmaster	09b2c7497743db875ed446da95123175dd00b1c8
Symbolica	e679e8c129ce15ec50c46ffb4ff4b59aeb7813b8
Compilers	Rust 1.98.1 · gfortran 15.3.0
Report tools	Python 3.14.7 · Typst 0.15.1
Reference libraries	Ninja 1.2.1 · OneLOop 3.6 · Collier 1.2.9

Case	MG5aMC generate command	Point SHA-256 prefix
qcd-1	`u u~ > z [virt=QCD]`	points
ew-1	`u d~ > w+ [virt=QED]`	points
qcd-2	`g g > t t~ [virt=QCD]`	points
ew-2	`e+ e- > mu+ mu- [virt=QED]`	points
qcd-3	`u u~ > z g g [virt=QCD]`	points
ew-3	`e+ e- > t t~ a [virt=QED]`	points
qcd-4	`u u~ > z z g g [virt=QCD]`	points
ew-4	`e+ e- > mu+ mu- a a [virt=QED]`	points

QCD cases use `loop_sm`; electroweak cases use `loop_qcd_qed_sm_Gmu`. Full dependency revisions, source archive and model hashes are in `provenance.json`. Full point/card hashes, per-run executable hashes when recorded, raw timings, Laurent coefficients, statuses and run logs remain with the benchmark inputs. `report-data.json` includes a SHA-256 manifest of the inputs used to build it.

Reproduce this validated report

```
source .ninjars-build/dev-env.sh
python tools/ninjars_report_v2.py DATASET --compile
```

Replace `DATASET` with the retained dataset path in `report-data.json`. Collection and correction specifications, CLI cards, all source roles, exact postimages, fresh/reused worker proof and raw measurements are bound by the dataset manifest. See `policy-v2/README.md` for the complete collection sequence. The renderer independently recomputes all numerical comparisons and confidence intervals. It rejects changed inputs and stale conclusions, and writes into a new directory so earlier reports remain intact. When diagnostic profiles are included, also pass `--profile-spec` with the retained profile manifest listed in the snapshot's input hashes.

Dataset dataset · policy policy · correction correction · representative proof proof.

Style attribution. Typography, margins, navy headings, pale notes, striped tables, provider cards and the logarithmic green–white–red ratio palette follow MadGraphTeam/pyamplicol's one-loop benchmark report. Its physics results and performance claims are not reused here.